

Report

Paper 1 Machine learning and AI in Bioinformatics

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Abstract

Goal of the project: Develop predictive models to classify metabolomics samples into Estrogen (E) and Estrogen + Progestin (E+P).

Main results of the project: The tuned Penalised Logistic Regression (PCLR) model achieved an accuracy of 0.67 and an AUC of 0.74, outperforming Random Forest.

Personal key learning: Preprocessing, feature selection, and dimensionality reduction had a greater impact on performance than model complexity in high-dimensional metabolomics datasets.

Estimated working hours: 12

Project evaluation: 2

Number of words: 1200

Keywords: Machine Learning, Metabolomics, Predictive Modelling

1 Scientific Background

Metabolomics datasets are typically high-dimensional and contain correlated variables. This makes machine learning challenging due to overfitting. PCA is commonly used to reduce dimensionality and improve model stability.

2 Goal

The goal was to classify Estrogen (E) vs Estrogen + Progestin (E+P) using machine learning and compare model performance.

3 Data and Preprocessing

The MTBLS136 metabolomics dataset contained 949 metabolite features measured across multiple biological samples. For this analysis, the dataset was restricted to a binary classification problem by selecting two relevant classes, resulting in a total of 668 samples. Preprocessing was performed to improve data quality and model reliability. This included feature selection based on the Peak Table, removal of features with excessive missing values, and KNN imputation to handle remaining missing observations. A log transformation was applied to reduce skewness and stabilise feature distributions, followed by standard scaling to ensure comparable feature magnitudes across variables. Finally, the processed dataset was divided into training and test sets using stratified sampling (80:20 split) to preserve the original class distribution and support unbiased model evaluation.

4 Methods

For evaluation purposes, the processed dataset was split into an 80 training set and a 20 stratified sampling to preserve class proportions. Two classification models were implemented using the scikit-learn library. First, a Random Forest (RF) classifier was trained as an ensemble tree-based method capable of capturing nonlinear relationships in the data. Hyperparameter tuning was performed using grid search with cross-validation to optimise model performance.

Second, a Principal Component Logistic Regression (PCLR) model was developed. Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset while retaining 95% of the variance. Logistic Regression was then trained on the transformed components, with hyperparameter tuning applied to the regularisation parameter.

Model performance was evaluated on the test set using classification accuracy and the Area Under the Receiver Operating Characteristic Curve (AUC). Additionally, bootstrap

resampling was performed to assess the stability and robustness of the model performance across multiple resampled datasets.

5 Results

Model	Train Accuracy	Test Accuracy	AUC
RF	1.000	0.638	0.705
RF Tuned	0.919	0.661	0.711
PCLR	1.000	0.647	0.715
PCLR Tuned	0.955	0.674	0.738

Table 1: Performance comparison of baseline and tuned models. Baseline models show overfitting, while tuning improves generalisation.

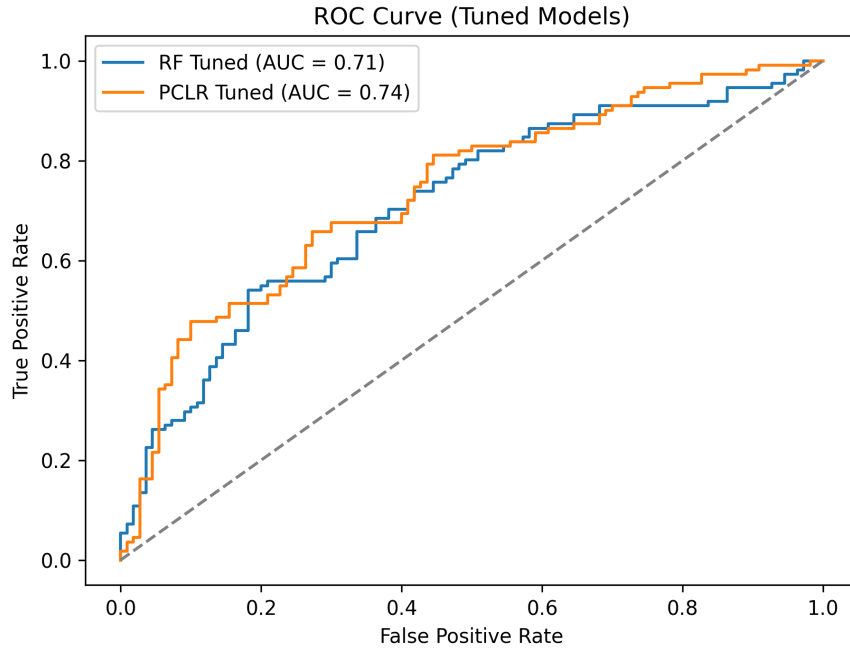


Figure 1: ROC curve comparing the tuned Random Forest and PCLR models. The PCLR model achieves a higher AUC, indicating better classification performance.

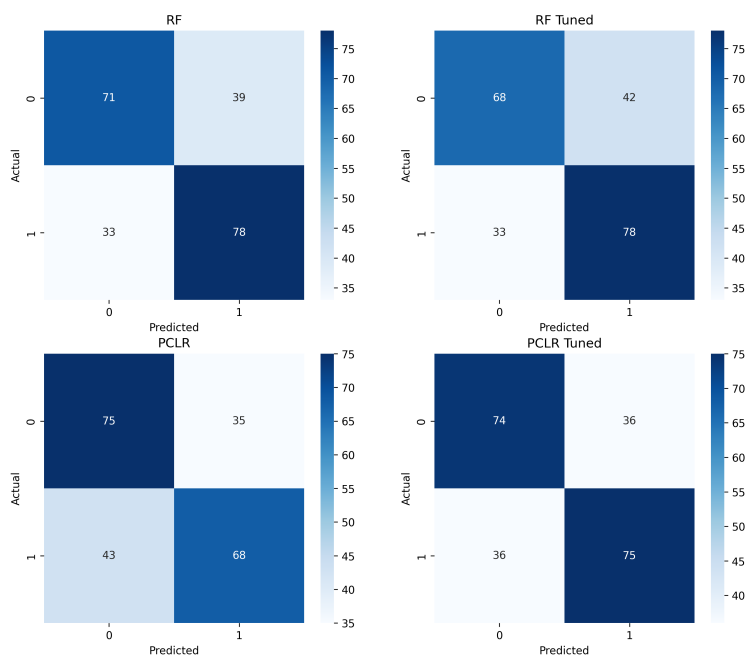


Figure 2: Confusion matrix illustrating classification performance of two classifier

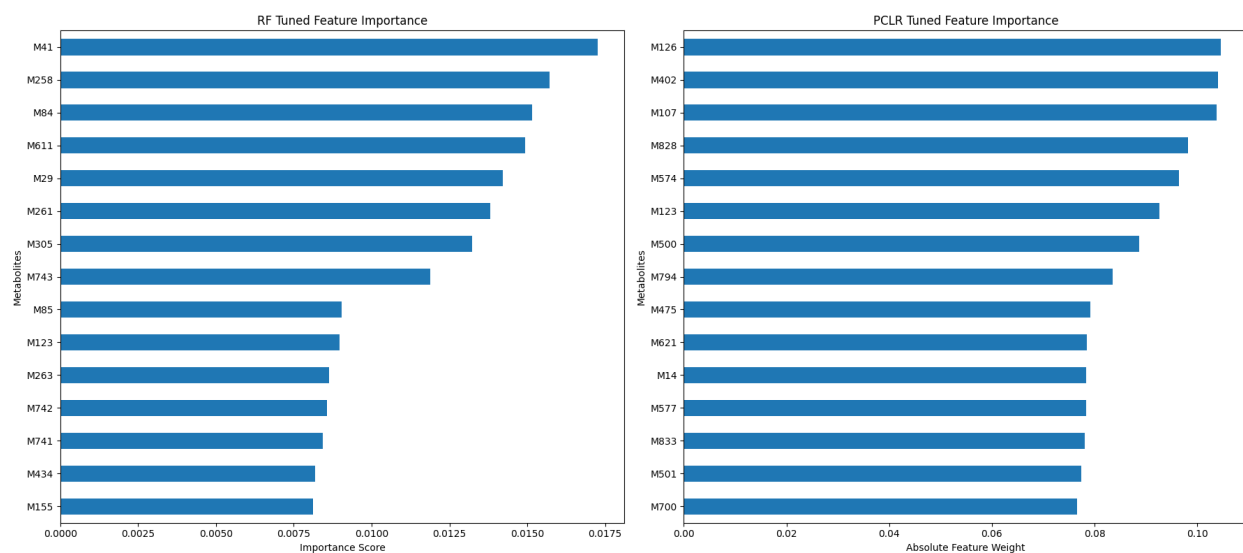


Figure 3: Comparison of top metabolite feature importance scores for the tuned Random Forest and tuned PCLR models.

6 Discussion

The results of this study compare Random Forest (RF) and Principal Component Logistic Regression (PCLR) for high-dimensional metabolomics data using accuracy, AUC, and

bootstrap validation. Both baseline models showed clear overfitting, achieving perfect training accuracy (1.00) but lower test accuracy (0.64 for RF and 0.65 for PCLR), indicating poor generalisation due to the high number of correlated features.

Hyperparameter tuning improved performance by reducing overfitting. The tuned RF model decreased training accuracy to 0.92 while improving test accuracy to approximately 0.70 and AUC to 0.71. However, the improvement remained moderate, suggesting sensitivity to noise. In contrast, the tuned PCLR model achieved the best performance, with training accuracy of 0.96, test accuracy of 0.73, and AUC of 0.74. This indicates better generalisation, largely due to PCA reducing dimensionality and removing redundant features.

These results align with the reference study, which reported AUC values in the range of 0.63–0.78 and found only small differences between models, with linear approaches performing comparably or slightly better. The results obtained here fall within this range, confirming consistency with expected behaviour.

Bootstrap validation further demonstrated that the tuned PCLR model is stable, with low variability in AUC across resampled datasets. This indicates that the model performance is robust and not dependent on a specific train-test split.

Although the tuned PCLR model slightly outperformed RF (0.73 vs 0.70 test accuracy), the difference is small, reinforcing that preprocessing and dimensionality reduction are more important than model complexity. However, limitations include the high feature-to-sample ratio and the restriction to binary classification. Future work could explore multiclass analysis and more robust validation techniques.

The findings show that both models identified important metabolite predictors, but with different patterns. RF relied on a broader range of features, reflecting its ability to capture complex nonlinear relationships. In contrast, PCLR placed greater emphasis on fewer metabolites due to its dimensionality reduction and coefficient-based structure. The overlap in some features suggests potentially robust biomarkers, while differences indicate that each model captures distinct predictive patterns in the metabolomic data. This overlap indicates that M123 is a relatively robust metabolomic biomarker for distinguishing Estrogen and Estrogen + Progestin samples, while the non-overlapping metabolites likely capture model-specific predictive patterns.

Overall, the findings show that PCLR is more effective for this task and confirm that simpler models combined with dimensionality reduction can achieve strong and reliable performance in metabolomics data analysis.

7 References

Smilde, A. K., Jansen, J. J., Hoefsloot, H. C. J., Lamers, R. J. A. N., van der Greef, J., Timmerman, M. E. (2005). ANOVA-simultaneous component analysis (ASCA): A new tool for analyzing designed metabolomics data. *Bioinformatics*, 21(13), 3043–3048. <https://doi.org/10.1093/bioinformatics/bti130>

8 Contributions

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